SARS-CoV infection
ACE2, soluble
S
S
S
ACE2, membrane-bound
active
ACE2-SARS-CoV interaction
ACE2, membrane-bound
active
ADAM17
ACE2
pulmonary fibrosis
viral RNA genome replication
HSPA5
HSPA5-Spike interaction
viral entry into host cell
early endosome
PACS2
ADAM17
PACS2:ADAM17 Interaction
nucleus
ATF6
DNAJC3
presence of dsRNA
EIF2AK2
EIF2AK4
ATP
GCN2:ATP
EIF2AK1
EIF2S1
EIF2S2
EIF2S3
GDP
EIF2-P:GDP
P
Met-tRNA
EIF2S1
EIF2S2
EIF2S3
GTP
Ternary Complex
EIF2B1
EIF2B2
EIF2B3
EIF2B4
EIF2B5
EIF2B
Translation initiation
DDIT3
DDIT3
&
XBP1
MAPK14
TRIB3
ATF4
ERO1A
PPP1R15A
DNAJC3
Persistant ER Stress
cleaved~CASP3
CASP3
cleaved~CASP4
CASP4
cleaved~CASP9
CASP9
CAPN1
Ca2+
CAPN1:Ca2+
CAPN1
high Ca2+ cytosolic concentration
release of ER Ca2+
XBP1
ERAD
protein ubiquitination and destruction
Ca2+
TMBIM6
ITPR
RYR1
ERO1A
endoplasmic reticulum
ERN1
BCL2L11
ERN1:BCL2L11
ERN1
BBC3
ERN1:BBC3
ERO1A
hyperoxidation
BAX
ERN1
BAX:ERN1
accumulation of misfolded protein in ER
ER Stress
Persistant ER Stress
Unfolded protein
retrograde transport from ER to cytosol
MFN2
endoplasmic reticulum organization
Apoptosis
unfolded protein response (UPR)
ER stress
endoplasmic reticulum calcium ion homeostasis
mitochondrion-endoplasmic reticulum membrane tethering
HSPA5
BAX
BAK1
BCL2
BCL2L1
Unfolded protein
Unfolded protein
ERN1
TRAF2
TRAF2:ERN1
BAK1
ERN1
BAK1:ERN1
Unfolded protein
EIF2AK3
EIF2AK3:EIF2AK3
ATF6
p20
PACS2
autophagy
VAPB
Ca2+ ER depletion
ITPR3
SIGMAR1
SIGMAR1:ITPR3
ITPR3
SIGMAR1
TRAF2
MFN2
MFN1
MFN1:MFN2
MFN2
MFN2
MFN2:MFN2
VAPB
RMDN3
VAPB:RMDN3
mitochondria-associated endoplasmic reticulum membrane
BCAP31
FIS1
FIS1:BCAP31
ITPR3
HSPA9
VDAC1
ITPR3:HSPA9:VDAC1
Ca2+ cytosolic concentration
RMDN3
FIS1
mitochondrion
mitochondrial fission
mitochondria fragmentation
MCU
SMDT1
SMDT1
MICU1
MICU2
MCUb
MICU1
MICU2
Mitochondrial calcium uniporter complex
mitochondrial matrix
mitochondrial outer membrane depolarization
Ca2+ mitochondrial concentration
mitochondrial calcium ion transmembrane transport
TRAF2
MAP3K5
BAX
BID
Cell survival
cell
CDK5
MAPK8
MAP3K4
MAP3K4
BCL2L11
Apoptosis
AP-1
PPP2CA
DDIT3
DDIT3
TRIB3
UPR
XBP1
TRIM13
FADD
MAP1LC3A
CASP8
K63
Ub
ATG5
CASP8:FADD:MAP1LC3A:SQSTM1:ATG5
Golgi apparatus
MBTPS1
MBTPS2
ATF6

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